01 / GNSS INTEGRITY

Is my GNSS usable?

Northwatch is a standalone prototype for a passive vessel trial. It records receiver evidence and flags candidate advisories for the watch and the debrief. Jamming and spoofing indicators can occur together.

Advisory	What the receiver may show	Watch response
Suspected jamming	Interference can degrade or block GNSS reception. A fix may become unreliable or disappear.	Use an approved independent fix. Fix loss alone does not identify jamming.
Suspected spoofing	The receiver may still output a plausible but false position. Record reported latitude, longitude and UTC.	Verify independently. Preserve the suspect coordinates for the debrief.
Monitor unavailable	Receiver data is missing or stale, or the dedicated helm link is down.	Treat integrity as unassessed and verify position by approved means.
No anomaly detected	Receiver diagnostics show no anomaly.	Independent confirmation comes from a timely fix under vessel procedure.

Where does the suspect position appear?

Plot the receiver-reported point as **Suspect GNSS position**. Plot an independent own-ship fix separately when available. Record UTC, source and uncertainty for each. Example (synthetic): GNSS reported 51.3500 N, 2.9100 E; independent fix 51.4000 N, 2.7500 E. Without a usable independent fix, record actual position and displacement as unknown. The reported point is where the receiver placed the vessel. It does not locate the interfering transmitter.

For the operating procedure, consult the vessel’s approved guidance and MCA MGN 719.

02 / INTEGRATION AND TRIAL

What we build and test

One outdoor multiband antenna, a Septentrio receiver, a standalone CloudGate recorder and a dedicated Ethernet link to an indoor helm display.

Component	Prototype configuration
Septentrio mosaic-go X5	Multiband receiver with SBF and AIM+ diagnostics. Capture RFStatus, ReceiverStatus and PVTGeodetic where supported. Verify antenna coverage, cable loss and the USB or compatible serial interface.
CloudGate and helm display	CG0124-12135 candidate: 450 MHz i.MX280, 128 MB RAM, 512 MB flash and two 10/100 Ethernet ports. Run a local recorder and browser UI. Use dedicated Ethernet and approved local power. Keep radios disabled for the initial trial. Confirm storage, firmware and startup on the actual unit.
Logging and replay	Record receiver messages, UTC and monotonic timestamps, data age, candidate advisories and crew markers. Rotate logs within a storage quota and export approved records locally.

Trial case	What to measure
Passive passage	Log coverage, gaps, bytes per hour, nuisance advisories and crew observations. Keep the initial trial silent.
Shielded RF test or replay	Test jamming and spoofing separately, including gradual spoofed drift. Measure time to advisory, missed events and position error against a known reference.
Receiver or cable disconnect	Measure time to unavailable and recovery. Confirm recorder logging continues while the helm link is down.
Independent fix	Compare suspect and independently observed positions with source, time and uncertainty. AIS can provide supporting evidence.

Agree acceptance thresholds before boarding. Verify boot-to-ready time, sustained logging and power on the assembled kit.